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United States Patent	12389329
Kind Code	B2
Date of Patent	August 12, 2025
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Sidelink communication with reduced power consumption

Abstract

Example embodiments relate to sidelink communication with reduced power consumption. In an embodiment, a terminal device may comprise at least one processor and at least one memory including computer program code. The at least one memory and the computer program code may be configured to, with the at least one processor, cause the terminal device to monitor sidelink transmission during a monitoring time-interval of a transmission period, and ignore a re-transmission time-interval subsequent to the monitoring time-interval in the transmission period in a case where no sidelink transmission to the terminal device is received in the monitoring time-interval.

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Appl. No.:	18/000271
Filed (or PCT Filed):	June 12, 2020
PCT No.:	PCT/CN2020/095902
PCT Pub. No.:	WO2021/248476
PCT Pub. Date:	December 16, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230209468 A1	Jun. 29, 2023

Publication Classification

Int. Cl.: H04W52/02 (20090101); H04W72/1263 (20230101); H04W72/25 (20230101)

U.S. Cl.:

CPC H04W52/0225 (20130101); H04W72/1263 (20130101); H04W72/25 (20230101);

Field of Classification Search

CPC: H04W (52/0225); H04W (72/1263); H04W (72/25); H04W (52/241); Y02D (30/70); H04L (1/1887); H04L (1/189); H04L (5/0044); H04B (7/0695)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage entry of International Application No. PCT/CN2020/095902, filed Jun. 12, 2020, entitled “SIDELINK COMMUNICATION WITH REDUCED POWER CONSUMPTION” which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(2) Exemplary embodiments described herein generally relate to communication technologies, and more particularly, to wireless communication devices, methods and systems for sidelink communication with reduced power consumption.

BACKGROUND

(3) Certain abbreviations that may be found in the description and/or in the figures are herewith defined as follows:

(4) TABLE-US-00001 D2D Device-to-Device gNB 5G Node-B LTE Long Term Evolution MAC Medium Access Control NR New Radio PS Public Safety PSCCH Physical Sidelink Control Channel PSFCH Physical Sidelink Feedback Channel PSSCH Physical Sidelink Shared Channel RLC Radio Link Control UE User Equipment V2X Vehicle-to-Everything VOIP Voice over IP

(5) Sidelink was firstly introduced in 3GPP Rel. 12 as an LTE feature for enabling device-to-device (D2D) communications, and it continuously evolved in Rels. 13/14/15 for various applications especially in vehicle-to-everything (V2X) scenarios. The first standard for NR based sidelink was established in Rel. 16, which also focused on V2X.

(6) Recently, enhancement to NR sidelink has been proposed with an aim of extending NR sidelink usage to public safety and commercial use cases. Since these use cases have requirements and traffic characteristics quite different from those in V2X, corresponding enhancements are needed on the top of NR V2X sidelink specified in Rel. 16. For example, in the public safety use cases, UEs are usually sensitive to energy consumption, and an efficient power saving mechanism is particularly desirable.

SUMMARY

(7) A brief summary of exemplary embodiments is provided below to provide basic understanding of some aspects of various embodiments. It should be noted that this summary is not intended to identify key features of essential elements or define scopes of the embodiments, and its sole purpose is to introduce some concepts in a simplified form as a preamble for a more detailed description provided below.

(8) In a first aspect, an example embodiment of a network device is provided. The network device may comprise at least one processor and at least one memory including computer program code. The at least one memory and the computer program code may be configured to, with the at least one processor, cause the network device to configure a terminal device with transmission periods for sidelink communication. The transmission periods may individually comprise a monitoring time-interval for at least a first transmission of a data packet and a re-transmission time-interval subsequent to the monitoring time-interval for a re-transmission of the data packet.

(9) In a second aspect, an example embodiment of a terminal device is provided. The terminal device may comprise at least one processor and at least one memory including computer program code. The at least one memory and the computer program code may be configured to, with the at least one processor, cause the terminal device to transmit a first transmission of a data packet to a receiving terminal device during a monitoring time-interval in a transmission period, and to transmit subsequent transmissions of the data packet to the receiving terminal device during the monitoring time-interval and/or a re-transmission time-interval subsequent to the monitoring time-interval in the transmission period.

(10) In a third aspect, an example embodiment of a terminal device is provided. The terminal device may comprise at least one processor and at least one memory including computer program code. The at least one memory and the computer program code may be configured to, with the at

least one processor, cause the terminal device to monitor sidelink transmission during a monitoring time-interval of a transmission period, and to ignore a re-transmission time-interval subsequent to the monitoring time-interval in the transmission period in a case where no sidelink transmission to the terminal device is received in the monitoring time-interval.

(11) In a fourth aspect, an example embodiment of a method implemented at a network device is provided. The method may comprise configuring a terminal device with transmission periods for sidelink communication. The transmission periods may individually comprise a monitoring time-interval for at least a first transmission of a data packet and a re-transmission time-interval subsequent to the monitoring time-interval for a re-transmission of the data packet.

(12) In a fifth aspect, an example embodiment of a method implemented at a terminal device is provided. The method may comprise transmitting a first transmission of a data packet to a receiving terminal device during a monitoring time-interval in a transmission period, and transmitting subsequent transmissions of the data packet to the receiving terminal device during the monitoring time-interval and/or a re-transmission time-interval subsequent to the monitoring time-interval in the transmission period.

(13) In a sixth aspect, an example embodiment of a method implemented at a terminal device is provided. The method may comprise monitoring sidelink transmission during a monitoring time-interval of a transmission period, and ignoring a re-transmission time-interval subsequent to the monitoring time-interval in the transmission period in a case where no sidelink transmission to the terminal device is received in the monitoring time-interval.

(14) In a seventh aspect, an example embodiment of a computer program product is provided. The computer program product may be embodied in at least one computer readable medium and comprise instructions. The instructions may, when executed by at least one processor in a network device, cause the network device to configure a terminal device with transmission periods for sidelink communication. The transmission periods may individually comprise a monitoring time-interval for at least a first transmission of a data packet, and a re-transmission time-interval subsequent to the monitoring time-interval for a re-transmission of the data packet.

(15) In an eighth aspect, an example embodiment of a computer program product is provided. The computer program product may be embodied in at least one computer readable medium and comprise instructions. The instructions may, when executed by at least one processor in a terminal device, cause the terminal device to transmit a first transmission of a data packet to a receiving terminal device during a monitoring time-interval in a transmission period, and to transmit subsequent transmissions of the data packet to the receiving terminal device during the monitoring time-interval and/or a re-transmission time-interval subsequent to the monitoring time-interval in the transmission period.

(16) In a ninth aspect, an example embodiment of a computer program product is provided. The computer program product may be embodied in at least one computer readable medium and comprise instructions. The instructions may, when executed by at least one processor in a terminal device, cause the terminal device to monitor sidelink transmission during a monitoring time-interval of a transmission period, and to ignore a re-transmission time-interval subsequent to the monitoring time-interval in the transmission period in a case where no sidelink transmission to the terminal device is received in the monitoring time-interval.

(17) Other features and advantages of the example embodiments of the present disclosure will also be apparent from the following description of specific embodiments when read in conjunction with the accompanying drawings, which illustrate, by way of example, the principles of example embodiments of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Some example embodiments will now be described, by way of non-limiting examples, with reference to the accompanying drawings.
- (2) FIG. 1 illustrates a schematic diagram of an example communication system in which embodiments of the present disclosure can be implemented.
- (3) FIG. 2 illustrates an interaction diagram of operations for sidelink communication according to some embodiments of the present disclosure.
- (4) FIG. 3 illustrates a schematic diagram of sidelink configuration for a receiving terminal device according to some embodiments of the present disclosure.
- (5) FIG. 4 illustrates a schematic diagram of sidelink configuration for a receiving terminal device according to some embodiments of the present disclosure.
- (6) FIGS. 5A, 5B, 5C illustrate examples of resource selection for sidelink transmissions according to some embodiments of the present disclosure.
- (7) FIGS. 6A-6B illustrate examples of resource selection for sidelink transmissions according to some embodiments of the present disclosure.
- (8) FIG. 7 illustrates a flow chart of a method implemented at a network device for sidelink communication according to some embodiments of the present disclosure.
- (9) FIG. 8 illustrates a flow chart of a method implemented at a terminal device for sidelink communication according to some embodiments of the present disclosure.
- (10) FIG. 9 illustrates a flow chart of a method implemented at a terminal device for sidelink communication according to some embodiments of the present disclosure.
- (11) FIG. 10 illustrates a schematic block diagram of a device according to some embodiments of the present disclosure.
- (12) Throughout the drawings, same or similar reference numbers indicate same or similar elements. A repetitive description on the same elements would be omitted.

DETAILED DESCRIPTION

- (13) Herein below, some example embodiments are described in detail with reference to the accompanying drawings. The following description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known circuits, techniques and components are shown in block diagram form to avoid obscuring the described concepts and features.
- (14) As used herein, the term “network device” refers to any suitable entities or devices that can provide cells or coverage, through which the terminal device can access the network or receive services. The network device may be commonly referred to as a base station. The term “base station” used herein can represent a node B (NodeB or NB), an evolved node B (eNodeB or eNB), or a gNB. The base station may be embodied as a macro base station, a relay node, or a low power node such as a pico base station or a femto base station. The base station may consist of several distributed network units, such as a central unit (CU), one or more distributed units (DUs), one or more remote radio heads (RRHs) or remote radio units (RRUs). The number and functions of these distributed units depend on the selected split RAN architecture.
- (15) As used herein, the term “terminal device” or “user equipment” (UE) refers to any entities or devices that can wirelessly communicate with the network devices or with each other. Examples of the terminal device can include a mobile phone, a mobile terminal (MT), a mobile station (MS), a subscriber station (SS), a portable subscriber station (PSS), an access terminal (AT), a computer, a wearable device, an on-vehicle communication device, a machine type communication (MTC) device, a D2D communication device, a V2X communication device, a sensor and the like. The term “terminal device” can be used interchangeably with a UE, a user terminal, a mobile terminal, a mobile station, or a wireless device.

(16) As used herein, the term “circuitry” may refer to one or more or all of the following: (a) hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry); and (b) combinations of hardware circuits and software, such as (as applicable): (i) a combination of analog and/or digital hardware circuit(s) with software/firmware; and (ii) any portions of hardware processor(s) with software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions); and (c) hardware circuit(s) and/or processor(s), such as a microprocessor(s) or a portion of a microprocessor(s), that requires software (e.g., firmware) for operation, but the software may not be present when it is not needed for operation. This definition of circuitry applies to all uses of this term in this application, including in any claims. As a further example, as used in the present disclosure, the term “circuitry” also covers an implementation of merely a hardware circuit or processor (or multiple processors) or portion of a hardware circuit or processor and its (or their) accompanying software and/or firmware. The term circuitry also covers, for example and if applicable to the particular claim element, a baseband integrated circuit or processor integrated circuit for a mobile device or a similar integrated circuit in server, a cellular network device, or other computing or network device.

(17) The term “include” and its variants are to be read as open-ended terms that mean “includes, but is not limited to.” The term “based on” is to be read as “based at least in part on.” The term “one embodiment” is to be read as “at least one embodiment.” The term “a further embodiment” is to be read as “at least a further embodiment.” Definitions related to other terms will be described in the following description.

(18) FIG. 1 illustrates a schematic diagram of an example communication system **100** in which exemplary embodiments of the present disclosure can be implemented. Referring to FIG. 1, the communication system **100**, which is a part of a communication network, includes a network device **120** and a plurality of terminal devices (hereinafter “UE”) **110-1**, **110-2**, **110-3** (collectively referred to as “UE **110**”) in a group **130**. The network device **120** may communicate with the UEs **110** on uplink (UL), downlink (DL) and/or supplementary uplink (SUL), and the UEs **110-1**, **110-2**, **110-3** may communicate with each other on sidelink. It would be understood that the UE group **130** may include a different number of UEs, for example 2, 4, 5 or more UEs, and the communication system **100** may include a different number of network devices, and/or a different number of terminal devices, and/or a different number of UE groups. For example, the communication system **100** may include a police public safety (PS) UE group and a fireman PS UE group, and an UE may belong to one or both of the groups. For sidelink communication between UEs, each UE **110** may act as a transmitting device sending sidelink transmissions to another UE(s), and as a receiving device receiving sidelink transmissions from another UE(s). Any one of the UEs **110** in the group **130** may also act as a group leader which periodically transmits and/or receives data to/from group members, or the group **130** may not include a group leader. For the sake of concision and clarity, herein below the terminal device **120** is described as a gNB, the UE **110-1** is described as a transmitting UE, the UE **110-2** as a receiving UE, and the UE **110-3** as an additional UE, but it would be appreciated that embodiments are not limited in any way thereto.

(19) It would be understood that some embodiments may be implemented when the UEs **110** are within coverage of the network device **120** as shown in FIG. 1, some embodiments may be implemented when the UEs **110** are out of coverage of the network device **120**, and some embodiments may also be implemented when some of the UEs **110** are within coverage of the network device **120** and the rest of the UEs **110** are out of coverage of the network device **120**.

(20) In public safety and commercial use scenarios, UEs are usually more energy sensitive than on-vehicle devices. According to a power consumption model revealed in 3GPP TR 36.746, UE in a receiving state consumes 40 to 100 times more power than that in a sleep or idle state. In the receiving state, UE needs to perform blind decoding of a physical sidelink control channel (PSCCH) in each slot of a sidelink resource pool for the UE so as not to miss any packet of interest.

Blind decoding operations consume substantially amount of power. Therefore, it would be desirable to reduce inefficient blind decoding and put UE more in the sleep or idle state in order to decrease power consumption.

(21) FIG. 2 illustrates an interaction diagram of operations for sidelink communication according to some embodiments of the present disclosure. The process shown in FIG. 2 can decrease power consumption of a receiving UE for sidelink communication while it would not cause missing of any sidelink transmission to the receiving UE. Referring to FIG. 2, at Block 1, an operation of configuring sidelink for the receiving UE 110-2 is performed. The operation may be performed at the gNB 120 as shown by a dashed arrow 1.1, at the transmitting (Tx) UE 110-1 as shown by a dashed arrow 1.2, at the receiving (Rx) UE 110-2 itself as shown by a dashed block 1.3, or at the additional UE 110-3 as shown by a dashed arrow 1.4. In some embodiments, the additional UE 110-3 may act as a leader UE in the group 130 comprising the UEs 110-1, 110-2 and 110-3. When sidelink for the Rx UE 110-2 is configured by the Tx UE 110-1, the Rx UE 110-2 itself or the additional UE 110-3, they may also report the sidelink configuration of the Rx UE 110-2 to the gNB 120.

(22) In particular, at the operation 1, the Rx UE 110-2 may be configured with transmission periods for sidelink communication, of which an example is shown in FIG. 3. Referring to FIG. 3, the transmission periods for sidelink communication each may include a monitoring interval and a re-transmission interval subsequent to the monitoring interval in the time domain. The monitoring interval may include one or more slots for at least a first transmission of a data packet to the Rx UE 110-2, and the re-transmission interval may include one or more slots for re-transmissions of the data packet to the Rx UE 110-2. It would be understood that slots for the transmission periods may be selected from a resource pool assigned to the Rx UE 110-2.

(23) In some embodiments, first m transmissions of the data packet to the Rx UE 110-2 may be transmitted during the monitoring interval, and remaining n transmissions of the data packet may be transmitted during the re-transmission interval, where m is an integer equal to or greater than one, and n is an integer equal to or greater than zero. The numbers m, n may be configured to the Rx UE 110-2 by for example a first parameter m and a second parameter n, respectively. The number m may be determined depending on quality of the channel between the Tx UE 110-1 and the Rx UE 110-2. If the channel quality is good, m may have a small value; if the channel quality is poor, m may have a large value.

(24) In some embodiments, the Rx UE 110-2 may be configured with identical monitoring intervals as other UEs in the same service group 130. For instance, UEs used by policemen belong to a police public safety (PS) group and they are configured with identical monitoring intervals (or transmission periods) so that each UE in the police PS group can receive sidelink transmissions from other UEs in the group; UEs used by firemen belong to a fire brigade PS group and they are configured with identical monitoring intervals (or transmission periods) so that each UE in the fire brigade PS group can receive sidelink transmissions from other UEs in the group. The monitoring intervals for the police PS group may not overlap with the monitoring intervals for the fire brigade PS group since UEs in the police PS group do not need to receive transmissions from UEs in the fire brigade PS group. In some embodiments, identical monitoring intervals may be configured for both the police PS group and the fire brigade PS group to facilitate inter-department communications.

(25) In some embodiments, a size of the monitoring interval may be determined based on transmitter density, the number m, channel busy ratio (CBR), and/or resource selection methods. For example, if many Tx UEs are sending data to the Rx UE 110-2, the number m has a large value, and/or the channel is busy, the monitoring interval may be configured with a large size. For another example, the monitoring interval may be configured with a larger size in a case of random resource selection than in a case of sensing-based resource selection.

(26) In some embodiments, a size of the transmission period may be determined based on services

provided by the sidelink communication. For instance, in a public safety use scenario, the most important service needed to be supported is VOIP (voice-over-IP), and a voice codec typically delivers a VoIP packet per 20 ms. Hence a suitable choice for the transmission period may be 20 ms or multiple times of 20 ms such as 40 ms, 60 ms, etc., as long as it is shorter than a latency requirement for the sidelink communication.

(27) FIG. 4 shows another example of the sidelink configuration for the Rx UE **110-2** according to some embodiments of the present disclosure, in which multiple consecutive transmission periods shown in FIG. 3 are configured as a long period. For instance, 5 transmission periods are configured as a long period in FIG. 4, and each long period includes 5 monitoring intervals and 5 re-transmission intervals. It would be understood that more or less transmission periods may be configured as a long period, as long as the long period is shorter than a latency requirement for the sidelink communication. For instance, when the transmission period is 20 ms, the long period of FIG. 4 has a size of 100 ms, which is shorter than a typical end-to-end latency requirement for the VOIP service, i.e., 200 ms.

(28) The long period may be inactive or active. The Rx UE **110-2** always monitors the first monitoring interval of a long period to determine if the long period is active or inactive. If no sidelink transmission is scheduled for the Rx UE **110-2** in the first monitoring interval, the Rx UE **110-2** determines that the long period is inactive and then it does not need to monitor subsequent monitoring intervals in the long period. On the other hand, if at least one sidelink transmission is scheduled for the Rx UE **110-2** in the first monitoring interval, the Rx UE **110-2** determines that the long period is active and it keeps monitoring every monitoring interval in the long period.

(29) Referring back to FIG. 2, at Block 2, sidelink transmissions are scheduled for the Rx UE **110-2** when the Tx UE **110-1** has a data packet to be transmitted to the Rx UE **110-2**. For example, when a Media Access Control layer of the Tx UE **110-1** receives from a higher layer a packet data unit (PDU) to be transmitted to the Rx UE **110-2**, the Tx UE **110-1** needs to schedule resources for transmissions of the data packet. In some embodiments, sidelink transmissions to the Rx UE **110-2** may be scheduled by the gNB **120**, as shown by a dashed block 2.1. For instance, the Tx UE **110-1** may send a scheduling request to the gNB **120** informing that it has one or more data packets to be transmitted to the Rx UE **110-2**. Then, the gNB **120** will select available resources for transmissions of the data packets and send a scheduling grant carried in Downlink Control Information (DCI) to the Tx UE **110-1** informing the resource selection. In some embodiments, the sidelink transmissions to the Rx UE **110-2** may be scheduled by the additional UE **110-3**, as shown by a dashed block 2.3. Similar to the gNB **120**, the additional UE **110-3** may inform the Tx UE **110-1** of resource selection carried in a PSSCH or a Sidelink Control Information (SCI). In some embodiments, the sidelink transmissions to the Rx UE **110-2** may be scheduled by the Tx UE **110-1** itself, as shown by a dashed block 2.2. Various resource selection methods may be applied, including but not limited to random resource selection or sensing-based resource selection. For the sensing-based resource selection, the Tx UE **110-1** may sense status of control and data channels used by transmitting UEs and select available resources for transmissions to the Rx UE **110-2**.

(30) FIGS. 5A, 5B, 5C illustrate examples of resource selection for sidelink transmissions according to some embodiments of the present disclosure. As discussed above, the Rx UE **110-2** is configured with periodical monitoring and re-transmission intervals. When a data packet arrives, at least a first transmission of the data packet has to be scheduled during the monitoring interval, and subsequent transmissions of the data packet could be scheduled during the monitoring interval and/or the re-transmission interval.

(31) FIG. 5A shows a case where a data packet to the Rx UE **110-2** arrives during a re-transmission interval. Referring to FIG. 5A, first two transmissions ($m=2$) of the data packet are scheduled during the next monitoring interval, and subsequent two transmissions ($n=2$) of the data packet are scheduled during the re-transmission interval corresponding to the monitoring interval for the first two transmissions. It would be understood that four transmissions of the data packet are shown

merely by way of example, and more or less transmissions may be scheduled for the data packet. In addition, the numbers (m, n) of transmissions scheduled for the monitoring interval and the re-transmission interval may also vary depending on channel state between the Tx UE **110-1** and the Rx UE **110-2** and/or configuration of the communication system **100**. For example, if the channel between the Tx UE **110-1** and the Rx UE **110-2** has good quality, the number (m) of transmissions scheduled during the monitoring interval may decrease to for example one, and the number (n) of transmissions scheduled during the re-transmission interval may increase to for example three. On the other hand, if the channel between the Tx UE **110-1** and the Rx UE **110-2** has poor quality, the number (m) of transmissions scheduled during the monitoring interval may increase to for example three, and the number (n) of transmissions scheduled during the re-transmission interval may decrease to for example one.

(32) It would also be understood that the four transmissions of the data packet may be scheduled one by one. For example, when the Tx UE **110-1** sends the first transmission to the Rx UE **110-2**, the Rx UE **110-2** may send a Hybrid Automatic Repeat reQuest (HARQ) feedback to the Tx UE **110-1** on a physical sidelink feedback channel (PSFCH). If the HARQ feedback is an Acknowledgement (ACK) indicating that the Rx UE **110-2** has successfully received the transmission, the Tx UE **110-1** would not schedule a re-transmission of the data packet to the Rx UE **110-2**. Instead, the Tx UE **110-1** would flush its HARQ buffer and start a new transmission. On the other hand, if the HARQ feedback is a negative acknowledgement (NACK) indicating that the Rx UE **110-2** did not receive the transmission, the Tx UE **110-1** would schedule a re-transmission of the data packet to the Rx UE **110-2**. The process may repeat until the Rx UE **110-2** successfully receives the data packet or the number of transmissions of the data packet reaches a predetermined maximum.

(33) In some embodiments, resources for the four transmissions may also be determined at one time. For example, the gNB **120** (Block **2.1**), the Tx UE **110-1** (Block **2.2**) or the additional UE **110-3** (Block **2.3**) may select resources for the four transmissions of the data packet at one time. In particular, resources for the first two transmissions are selected from the monitoring interval and resources for the subsequent two transmissions are selected from the re-transmission interval corresponding to the monitoring interval. Then, the Tx UE **110-1** will transmit the four transmissions using the selected resources, and the Rx UE **110-2** will send an HARQ feedback to the Tx UE **110-1** after the last transmission.

(34) FIGS. **5B** and **5C** show cases where a data packet to the Rx UE **110-2** arrives during a monitoring interval. In some embodiments, as shown in FIG. **5B**, resources for first m (m=2 in FIGS. **5A-5C**) transmissions of the data packet may be selected from a next monitoring interval, but not from the current monitoring interval. In some embodiments, as shown in FIG. **5C**, resources for first m transmissions of the data packet may be selected from the current monitoring interval. Resources for the subsequent n (n=2 in FIGS. **5A-5C**) transmissions of the data packet may be selected from the re-transmission interval corresponding to the monitoring interval for the first m transmissions. Compared with the example shown in FIG. **5B**, the example shown in FIG. **5C** has a shorter latency, but it may increase collision probability of the transmissions. Hence, sensing-based resource selection is preferred. The sensing-based resource selection may be performed to check whether there are enough good transmission resources in the current monitoring interval. If not, the transmission resources may be selected from a next monitoring interval. It would also be understood that if the next monitoring interval shown in FIGS. **5A-5B** do not have enough good transmission resources, the first m transmissions may be scheduled to a next monitoring interval.

(35) FIGS. **6A-6B** illustrate other examples of resource selection for sidelink transmissions according to some embodiments of the present disclosure. In the examples of FIGS. **6A-6B**, the Rx UE **110-2** is configured with long transmission periods as shown in FIG. **4**, and each long period comprises multiple (5 in FIGS. **4, 6A-6B**) transmission periods shown in FIG. **3**. Referring to FIG. **6A**, when one or more (2 shown in FIG. **6A**) data packets to the Rx UE **110-2** arrive at an

inactivate long period in which no transmission to the Rx UE **110-2** was scheduled for the first monitoring interval of the inactivate long period, first m transmissions of at least one of the one or more data packets are scheduled during the first monitoring interval of the next long period. That is to say, the next long period would be an active long period. If the first monitoring interval of the next long period has enough good transmission resources, all of the one or more data packets arrived during the preceding inactive long period may have their first m transmissions scheduled during the first monitoring interval of the next long period. When a data packet to the Rx UE **110-2** arrives during the active long period in which the first monitoring interval is scheduled for transmissions to the Rx UE **110-2**, first m transmissions of the data packet would be scheduled during a current or next available monitoring interval, either in the current long period or in a next long period. Although not shown in FIG. **6A**, subsequent n transmissions of each data packet would be scheduled during the re-transmission interval corresponding to the monitoring interval for the first m transmissions of the data packet.

(36) FIG. **6B** shows a case where a data packet arrives during a first monitoring interval of a long transmission period (the first long period in FIG. **6B**) in which no transmission to the Rx UE **110-2** has been scheduled during the first monitoring interval of the long transmission period. In some embodiments, a first transmission of the data packet may be scheduled during the first monitoring interval of the long transmission period. Thus, the long transmission period becomes an active long period, and the Rx UE **110-2** will monitor each monitoring interval in the active long period. In some embodiments, the first transmission of the data packet may also be scheduled during a first monitoring time-interval of a next long transmission period. Thus, the current long transmission period becomes inactive, while the next long transmission period becomes active.

(37) Referring back to FIG. **2**, at Operation **3**, the Tx UE **110-1** transmits the data packet to the Rx UE **110-2** using the resources selected in Operation **2**, and at Operation **4**, the Rx UE **110-2** receives the sidelink transmissions. In particular, the Rx UE **110-2** monitors a monitoring interval of a transmission period. For example, the Rx UE **110-2** performs blind decoding of a physical sidelink control channel (PSCCH) in each slot of the monitoring interval and may further try to decode a corresponding physical sidelink shared channel (PSSCH) based on the decoded PSCCH. If a PSCCH part of one or more transmissions of a data packet to the Rx UE **110-2** is correctly received/decoded during the monitoring interval and a corresponding PSSCH part of the transmissions of the data packet is not correctly received/decoded, the Rx UE **110-2** will further decode subsequent transmission(s) of the data packet at resources in the corresponding re-transmission interval indicated in the previous transmission(s) of the data packet. In some embodiments, the Rx UE **110-2** may decode subsequent transmission(s) of the data packet at resources in the corresponding re-transmission interval indicated in the previous transmission(s) of the data packet if the PSCCH part of the previous transmission(s) is correctly decoded and the decoding of the PSSCH part of the previous transmission(s) is not performed, and the Rx UE **110-2** may combine all the transmissions to determine the data packet, thereby improving reliability of the communication. Referring to FIGS. **5A-5C**, a previous transmission may indicate resources of following two transmissions. Thus the Rx UE **110-2** may directly decode transmissions in the re-transmission interval, avoiding blind decoding of each slot in the re-transmission interval. If no transmission to the Rx UE **110-2** is received in the monitoring interval, or a transmission of a data packet to the Rx UE **110-2** is received in the monitoring interval and both the PSCCH part and the PSSCH part of the transmission are correctly decoded, the Rx UE **110-2** will ignore the re-transmission interval corresponding to the monitoring interval. In other words, the Rx UE **110-2** may be set in a sleep/idle mode with its receiver circuitry switched off. In this way, the Rx UE **110-2** may skip receiving/decoding the whole re-transmission interval or avoid receiving/decoding most slots in the re-transmission interval. Compared with a case of blind decoding all slots, the process of FIG. **2** allows for a significant reduction of power consumption for the sidelink communication.

(38) When the Rx UE **110-2** is configured with long periods as shown in FIG. **4**, it always monitors

the first monitoring interval in each long period. The Rx UE **110-2** turns on its receiving circuitry in the first monitoring interval of each long period and receives/decodes PSCCHs/PSSCHs in the interval (see ‘on’ and ‘off’ indications in FIG. 4). If at least one transmission is received in the first monitoring interval of the long period, the Rx UE **110-2** considers this long period active and keeps monitoring each subsequent monitoring interval in the long period (as for the second long transmission period in FIG. 4). If no transmission is received in the first monitoring interval of the long period, the Rx UE **110-2** considers this long period inactive and ignores all subsequent transmission periods in the long period (as for the first long transmission period in FIG. 4). This receiving mechanism can further reduce power consumption of the Rx UE **110-2** since there are substantial amount of silence time between talk spurts of VoIP data traffic, and it can also meet the latency requirement for the VOIP service.

(39) FIG. 7 illustrates a flow chart of a method **200** implemented at a network device for example the gNB **120** for sidelink communication according to some embodiments of the present disclosure. Referring to FIG. 7, the method **200** may comprise a step **S210** of configuring a terminal device (for example, the Rx UE **110-2**) with transmission periods for sidelink communication. As discussed above with reference to FIG. 3, the transmission period may comprise a monitoring interval for at least a first transmission of a data packet to the Rx UE **110-2** and a re-transmission interval subsequent to the monitoring interval for a re-transmission of the data packet.

(40) In some embodiments, optionally, the method **200** may further comprise a step **S220** of configuring the Rx UE **110-2** with parameters *m*, *n*. The first parameter *m* indicates that first *m* transmissions of the data packet to the Rx UE **110-2** would be scheduled during the monitoring interval, and the second parameter *n* indicates that subsequent *n* transmissions of the data packet would be scheduled during the re-transmission interval subsequent to the monitoring interval.

(41) In some embodiments, optionally, the method **200** may further comprise a step **S230** of configuring the Rx UE **110-2** with long periods, each including a plurality of transmission periods described above.

(42) In some embodiments, optionally, the method **200** may further comprise a step **S240** of scheduling a first transmission of a data packet to the Rx UE **110-2** during the monitoring interval of a transmission period and subsequent transmissions of the data packet to the Rx UE **110-2** during the monitoring interval and/or the re-transmission interval of the transmission period. For example, if the Rx UE **110-2** is configured with the first and second parameters *m*, *n*, first *m* transmissions of the data packet may be scheduled during the monitoring interval, and subsequent *n* transmissions of the data packet may be scheduled during the re-transmission interval subsequent to the monitoring interval. In some embodiments where the Rx UE **110-2** is configured with long periods as shown in FIG. 4, if the data packet arrives during an inactive long period in which no transmission to the Rx UE **110-2** was scheduled for the first monitoring interval of the long period, the first transmission of the data packet may be scheduled during the first monitoring interval in a next long period. If the data packet arrives during an active long period in which at least one transmission to the Rx UE **110-2** was scheduled for the first monitoring interval of the long period, the first transmission of the data packet may be scheduled during a current or next available monitoring interval in either the current active long period or a next long period. If the data packet arrives during a first monitoring interval of a long transmission period, the first transmission of the data packet may be scheduled during the first monitoring interval of the current long transmission period or during a first monitoring interval of a next long transmission period. Subsequent transmissions of the data packet may be scheduled during the monitoring interval for the first transmission and/or the re-transmission interval corresponding to the monitoring interval.

(43) FIG. 8 illustrates a flow chart of a method **300** implemented at a terminal device for example the Tx UE **110-1** for sidelink communication according to some embodiments of the present disclosure. Referring to FIG. 8, the method **300** may optionally comprise a step **S310a** of receiving resource selection for sidelink transmissions of a data packet to the Rx UE **110-2** from the gNB **120**

or the additional UE **110-3**, or a step **S310b** of selecting resources for transmissions of the data packet to the Rx UE **110-2**. As discussed above, resources for the first transmission of the data packet to the Rx UE **110-2** may be selected from a monitoring interval of a transmission period, and resources for subsequent transmissions of the data packet to the Rx UE **110-2** may be selected from the monitoring interval and/or a subsequent re-transmission interval in the transmission period. In particular, when the data packet arrives during a monitoring interval, resources for the first transmission of the data packet may be selected from a current or next available monitoring interval. When the data packet arrives during a re-transmission interval, resources for the first transmission of the data packet may be selected from a next available monitoring interval. In some embodiments where the Rx UE **110-2** is configured with long periods as shown in FIG. **4**, if the data packet arrives during an inactive long period in which no transmission to the Rx UE **110-2** was scheduled during the first monitoring interval of the long period, resources for the first transmission of the data packet may be selected from the first monitoring interval in a next long period. If the data packet arrives during an active long period in which at least one transmission to the Rx UE **110-2** was scheduled for the first monitoring interval of the long period, resources for the first transmission of the data packet may be selected from a current or next available monitoring interval in either the current active long period or a next long period. If the data packet arrives during a first monitoring interval of a long transmission period, resources for the first transmission of the data packet may be selected from the first monitoring interval of the current long transmission period or from a first monitoring interval of a next long transmission period. Resources for subsequent transmissions of the data packet may be selected from the monitoring interval for the first transmission and/or the re-transmission interval corresponding to the monitoring interval.

(44) In some embodiments, before the step **S310b** of selecting resources for transmissions of the data packet, the method **300** may further comprise a step **S320** of receiving sidelink configuration for the Rx UE **110-2** from the gNB **120**, the Rx UE **110-2** or the additional UE **110-3**. As discussed above, the sidelink configuration for the Rx UE **110-2** may comprise monitoring intervals, re-transmission intervals, transmission periods including the monitoring and re-transmission intervals, long periods including multiple transmission periods, and/or parameters *m*, *n* configured for the Rx UE **110-2**.

(45) Then at a step **S330**, transmissions of the data packet are transmitted to the Rx UE **110-2** by using the resource selection received from another device in the step **S310a** or determined at the Tx UE **110-2** in the step **S310b**. In particular, the first transmission of the data packet is transmitted during the monitoring interval, and subsequent transmissions of the data packet are transmitted during the monitoring interval and/or the re-transmission interval subsequent to the monitoring interval.

(46) FIG. **9** illustrates a flow chart of a method **400** implemented at a terminal device for example the Rx UE **110-2** for sidelink communication according to some embodiments of the present disclosure. Referring to FIG. **9**, the method **400** may optionally comprise a step **S410** of applying sidelink configuration at the Rx UE **110-2**. The sidelink configuration may be received from another device for example the gNB **120**, the Tx UE **110-1** or the additional UE **110-3**, or it may be determined at the Rx UE **110-2**. As discussed above, the sidelink configuration may comprise monitoring intervals, re-transmission intervals, transmission periods including the monitoring and re-transmission intervals, long periods including multiple transmission periods, and/or parameters *m*, *n* configured for the Rx UE **110-2**.

(47) Then at a step **S420**, the Rx UE **110-2** monitors sidelink transmissions during a monitoring interval of a transmission period. For example, the Rx UE **110-2** may perform blind decoding of a sidelink control channel in each slot of the monitoring interval. The Rx UE **110-2** may also decode a sidelink shared channel based on the decoded sidelink control channel.

(48) If it is determined at a step **S430** that at least one transmission of a data packet to the Rx UE **110-2** is received at the monitoring interval and a PSCCH part of the transmission is correctly

decoded while a corresponding PSSCH part is not correctly decoded or the decoding of the corresponding PSSCH part is not performed, then at a step **S440**, the Rx UE **110-2** may monitor subsequent transmissions of the data packet in a re-transmission interval subsequent to the monitoring interval. For example, the Rx UE **110-2** may directly decode the subsequent transmissions at resources indicated in the previous transmissions received in the monitoring interval. Optionally, if the Rx UE **110-2** is configured with long periods and the monitoring interval monitored at the step **S420** is a first monitoring interval in a long period, the method **400** may further comprise a step **S450** of monitoring subsequent monitoring intervals in the long period.

(49) If it is determined at the step **S430** that no transmission to the Rx UE **110-2** is received during the monitoring interval, or a transmission of a data packet to the Rx UE **110-2** is received during the monitoring interval and both a PSCCH part and a PSSCH part of the transmission are correctly decoded, then at a step **S460**, the Rx UE **110-2** will ignore a re-transmission interval subsequent to the monitoring interval in the transmission period. For example, the Rx UE **110-2** may switch off its receiver circuitry to save power during the re-transmission interval. Optionally, if the Rx UE **110-2** is configured with long periods and the monitoring interval monitored at the step **S420** is a first monitoring interval in a long period, the Rx UE **110-2** may further ignore subsequent transmission periods in the long period. For example, the Rx UE **110-2** may switch off its receiver circuitry to save power during all the monitoring intervals of the long period but the first, and during all the re-transmission intervals of the long period. It will further reduce power consumption for the sidelink communication.

(50) The methods **200**, **300**, **400** are briefly described herein as details thereof have been described when discussing the operations shown in FIG. **2**. It would be apparent that the description relating to the operations shown in FIG. **2** are also applicable to the methods **200**, **300** and **400** discussed herein.

(51) FIG. **10** illustrates a schematic block diagram of a device **500** according to some embodiments of the present disclosure. The device **500** may be implemented at the network device **120** or at any one of the UEs **110** to perform any one of the operations and/or methods discussed above.

(52) Referring to FIG. **10**, the device **500** may include one or more processors **511**, one or more memories **512** and one or more transceivers **513** interconnected through one or more buses **514**. The one or more buses **514** may be address, data, or control buses, and may include any interconnection mechanism such as series of lines on a motherboard or integrated circuit, fiber, optics or other optical communication equipment, and the like. Each of the one or more transceivers **513** may comprise a receiver and a transmitter, which are connected to one or more antennas **516**. The device **500** may wirelessly communicate with other devices through the one or more antennas **516**. For example, the device **500** may be implemented at a terminal device and it may communicate with other terminal devices on sidelinks and/or with one or more network devices on uplinks and downlinks. The one or more memories **512** may include computer program code **515**. The one or more memories **512** and the computer program code **515** may be configured to, when executed by the one or more processors **511**, cause the device **500** to perform operations and/or methods as described above.

(53) The one or more processors **511** may be of any appropriate type that is suitable for the local technical network, and may include one or more of general purpose processors, special purpose processor, microprocessors, a digital signal processor (DSP), one or more processors in a processor based multi-core processor architecture, as well as dedicated processors such as those developed based on Field Programmable Gate Array (FPGA) and Application Specific Integrated Circuit (ASIC). The one or more processors **511** may be configured to control other elements of the UE/network device and operate in cooperation with them to implement the procedures discussed above.

(54) The one or more memories **512** may include at least one storage medium in various forms, such as a volatile memory and/or a non-volatile memory. The volatile memory may include but not

limited to for example a random access memory (RAM) or a cache. The non-volatile memory may include but not limited to for example a read only memory (ROM), a hard disk, a flash memory, and the like. Further, the one or more memories 512 may include but not limited to an electric, a magnetic, an optical, an electromagnetic, an infrared, or a semiconductor system, apparatus, or device or any combination of the above.

(55) It would be understood that blocks shown in FIGS. 1-10 may be implemented in various manners, including software, hardware, firmware, or any combination thereof. In some embodiments, one or more blocks may be implemented using software and/or firmware, for example, machine-executable instructions stored in the storage medium. In addition to or instead of machine-executable instructions, parts or all of the blocks in FIGS. 1-10 may be implemented, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-Programmable Gate Arrays (FPGAs), Application-Specific Integrated Circuits (ASICs), Application-Specific Standard Products (ASSPs), System-on-Chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), etc.

(56) Some exemplary embodiments further provide computer program code or instructions which, when executed by one or more processors, may cause a device or apparatus to perform the procedures described above. The computer program code for carrying out procedures of the exemplary embodiments may be written in any combination of one or more programming languages. The computer program code may be provided to one or more processors or controllers of a general purpose computer, special purpose computer, or other programmable data processing apparatus, such that the program code, when executed by the processor or controller, cause the functions/operations specified in the flowcharts and/or block diagrams to be implemented. The program code may execute entirely on a machine, partly on the machine, as a stand-alone software package, partly on the machine and partly on a remote machine or entirely on the remote machine or server.

(57) Some exemplary embodiments further provide a computer program product embodied in a computer readable medium comprising the computer program code or instructions. The computer readable medium may be any tangible medium that may contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device. The machine readable medium may be a machine readable signal medium or a machine readable storage medium. A machine readable medium may include but is not limited to an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples of the machine readable storage medium would include an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing.

(58) Further, while operations are depicted in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Likewise, while several specific implementation details are contained in the above discussions, these should not be construed as limitations on the scope of the present disclosure, but rather as descriptions of features that may be specific to particular embodiments. Certain features that are described in the context of separate embodiments may also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment may also be implemented in multiple embodiments separately or in any suitable sub-combination.

(59) Although the subject matter has been described in a language that is specific to structural

features and/or method actions, it is to be understood the subject matter defined in the appended claims is not limited to the specific features or actions described above. On the contrary, the above-described specific features and actions are disclosed as an example of implementing the claims.

Claims

1. A terminal device comprising: at least one processor; and at least one memory including computer program code, the at least one memory and the computer program code configured to, with the at least one processor, cause the terminal device to: apply sidelink configuration at the terminal device, the sidelink configuration being determined by the following: a network device, a transmitting terminal device sending the sidelink transmission to the terminal device, an additional terminal device in sidelink communication with the terminal device, wherein the sidelink configuration indicates a monitoring time-interval, a re-transmission time-interval; a transmission period including the monitoring time-interval and the re-transmission time-interval; a long transmission period including a plurality of transmission periods; a first parameter m indicating that first m transmissions of a data packet are allowed to be transmitted to the terminal device during the monitoring time-interval, where the first parameter m is an integer equal to or greater than one; and a second parameter n indicating that subsequent n transmissions of the data packet are allowed to be transmitted to the terminal device during the re-transmission time-interval, where the second parameter n is an integer equal to or greater than zero; monitor sidelink transmission during the monitoring time-interval of the transmission period; determine that a control channel part of one or more transmissions of the data packet to the terminal device is correctly received during the monitoring time-interval and a corresponding data channel part is correctly received during the monitoring time-interval; and based on determining that the control channel part of the one or more transmissions of the data packet to the terminal device is correctly received during the monitoring time-interval and the corresponding data channel part is correctly received during the monitoring time-interval; ignore the re-transmission time-interval subsequent to the monitoring time-interval; and ignore subsequent period in the long transmission period.
2. The terminal device of claim 1 wherein monitoring sidelink transmission during a monitoring time-interval comprises performing blind decoding of a sidelink control channel in the monitoring time-interval.
3. The terminal device of claim 2, wherein the at least one memory and the computer program code are further configured to, with the at least one processor, cause the terminal device to: apply a second sidelink configuration at the terminal device; based on the second sidelink configuration, monitor sidelink transmission during a second monitoring time-interval of a second transmission period; determine that a transmission of a second data packet to the terminal device is received at the second monitoring time-interval and a physical sidelink control channel (PSCCH) part of the transmission of the second data packet is correctly decoded while a corresponding physical sidelink shared channel (PSSCH) part is not correctly decoded; and based on determining that the transmission of the second data packet to the terminal device is received at the second monitoring time-interval and the PSCCH part of the transmission of the second data packet is correctly decoded while the corresponding PSSCH part is not correctly decoded, monitor subsequent transmissions of the second data packet in a second re-transmission interval subsequent to the second monitoring interval.
4. A system comprising: a terminal device; at least one processor; and at least one memory including computer program code, the at least one memory and the computer program code configured to, with the at least one processor, cause the terminal device to: apply sidelink configuration at the terminal device, the sidelink configuration being determined by the following: a network device, a transmitting terminal device sending the sidelink transmission to the terminal device, an additional terminal device in sidelink communication with the terminal device, wherein

the sidelink configuration indicates a monitoring time-interval, a re-transmission time-interval; a transmission period including the monitoring time-interval and the re-transmission time-interval; a long transmission period including a plurality of transmission periods; a first parameter m indicating that first m transmissions of a data packet are allowed to be transmitted to the terminal device during the monitoring time-interval, where the first parameter m is an integer equal to or greater than one; and a second parameter n indicating that subsequent n transmissions of the data packet are allowed to be transmitted to the terminal device during the re-transmission time-interval, where the second parameter n is an integer equal to or greater than zero; monitor sidelink transmission during the monitoring time-interval of the transmission period; determine that a control channel part of one or more transmissions of the data packet to the terminal device is correctly received during the monitoring time-interval and a corresponding data channel part is correctly received during the monitoring time-interval; and based on determining that the control channel part of the one or more transmissions of the data packet to the terminal device is correctly received during the monitoring time-interval and the corresponding data channel part is correctly received during the monitoring time-interval: ignore the re-transmission time-interval subsequent to the monitoring time-interval; and ignore subsequent period in the long transmission period.

5. The system of claim 4, wherein the at least one memory and the computer program code are further configured to, with the at least one processor, cause the terminal device to: apply a second sidelink configuration at the terminal device; based on the second sidelink configuration, monitor sidelink transmission during a second monitoring time-interval of a second transmission period; determine that a transmission of a second data packet to the terminal device is received at the second monitoring time-interval and a physical sidelink control channel (PSCCH) part of the transmission of the second data packet is correctly decoded while a corresponding physical sidelink shared channel (PSSCH) part is not correctly decoded; and based on determining that the transmission of the second data packet to the terminal device is received at the second monitoring time-interval and the PSCCH part of the transmission of the second data packet is correctly decoded while the corresponding PSSCH part is not correctly decoded, monitor subsequent transmissions of the second data packet in a second re-transmission interval subsequent to the second monitoring interval.

6. A method comprising: applying, by a terminal device, sidelink configuration at the terminal device, the sidelink configuration being determined by the following: a network device, a transmitting terminal device sending the sidelink transmission to the terminal device, an additional terminal device in sidelink communication with the terminal device, wherein the sidelink configuration indicates a monitoring time-interval, a re-transmission time-interval; a transmission period including the monitoring time-interval and the re-transmission time-interval; a long transmission period including a plurality of transmission periods; a first parameter m indicating that first m transmissions of a data packet are allowed to be transmitted to the terminal device during the monitoring time-interval, where the first parameter m is an integer equal to or greater than one; and a second parameter n indicating that subsequent n transmissions of the data packet are allowed to be transmitted to the terminal device during the re-transmission time-interval, where the second parameter n is an integer equal to or greater than zero; monitoring, by the terminal device, sidelink transmission during the monitoring time-interval of the transmission period; determining, by the terminal device, that a control channel part of one or more transmissions of the data packet to the terminal device is correctly received during the monitoring time-interval and a corresponding data channel part is correctly received during the monitoring time-interval; and based on determining that the control channel part of the one or more transmissions of the data packet to the terminal device is correctly received during the monitoring time-interval and the corresponding data channel part is correctly received during the monitoring time-interval: ignoring, by the terminal device, the re-transmission time-interval subsequent to the monitoring time-interval; and ignoring, by the terminal device, subsequent period in the long transmission period.

7. The method of claim 6, further comprising: applying, by the terminal device, a second sidelink configuration at the terminal device; based on the second sidelink configuration, monitor, by the terminal device, sidelink transmission during a second monitoring time-interval of a second transmission period; determine, by the terminal device, that a transmission of a second data packet to the terminal device is received at the second monitoring time-interval and a physical sidelink control channel (PSCCH) part of the transmission of the second data packet is correctly decoded while a corresponding physical sidelink shared channel (PSSCH) part is not correctly decoded; and based on determining that the transmission of the second data packet to the terminal device is received at the second monitoring time-interval and the PSCCH part of the transmission of the second data packet is correctly decoded while the corresponding PSSCH part is not correctly decoded, monitor, by the terminal device, subsequent transmissions of the second data packet in a second re-transmission interval subsequent to the second monitoring interval.
